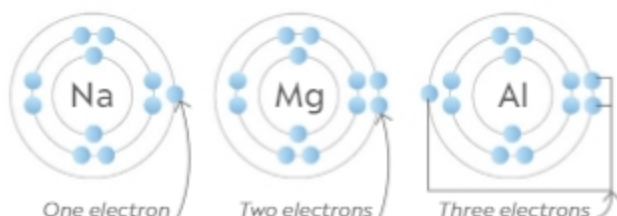


Periods

As you move across a period, the atoms of each element have the same number of shells. However, as you move across the period, the number of electrons in the outer shell increases. In the examples below, each element has three shells. The different numbers of electrons in the outer shells mean that these elements have differences in their chemical properties.



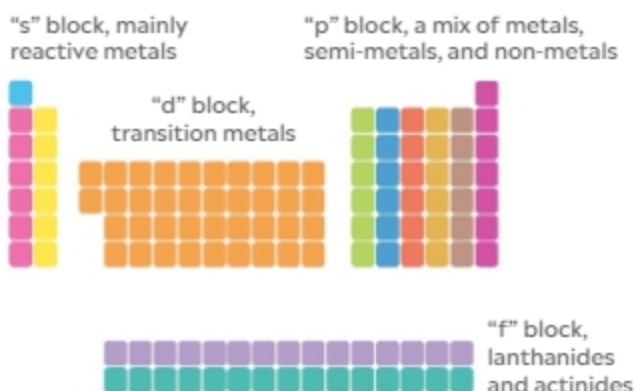
Sodium
A sodium atom has eleven electrons in total, with one in its outer shell.

Magnesium
A magnesium atom has twelve electrons in total, with two in its outer shell.

Aluminum
An aluminum atom has thirteen electrons in total, with three in its outer shell.

Blocks

Within the periodic table there are larger collections of elements known as blocks. Three of these blocks—the “s” block, the “d” block, and the “f” block—contain elements that have many broad similarities with one another, but the “p” block contains a much more diverse set of elements.



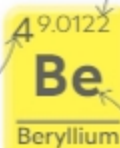
					18 2 4.0026 He Helium
13 5 10.811 B Boron	14 6 12.011 C Carbon	15 7 14.007 N Nitrogen	16 8 15.999 O Oxygen	17 9 18.998 F Fluorine	10 20.180 Ne Neon
13 26.982 Al Aluminum	14 28.086 Si Silicon	15 30.974 P Phosphorus	16 32.065 S Sulfur	17 35.453 Cl Chlorine	18 39.948 Ar Argon
31 69.723 Ga Gallium	32 72.64 Ge Germanium	33 74.922 As Arsenic	34 78.96 Se Selenium	35 79.904 Br Bromine	36 83.80 Kr Krypton
49 114.82 In Indium	50 118.71 Sn Tin	51 121.76 Sb Antimony	52 127.60 Te Tellurium	53 126.90 I Iodine	54 131.29 Xe Xenon
81 204.38 Tl Thallium	82 207.2 Pb Lead	83 208.98 Bi Bismuth	84 (209) Po Polonium	85 (210) At Astatine	86 (222) Rn Radon
113 284 Nh Nihonium	114 289 Fl Flerovium	115 288 Mc Moscovium	116 293 Lv Livermorium	117 292 Ts Tennessine	118 294 Og Oganesson
66 162.50 Dy Dysprosium	67 164.93 Ho Holmium	68 167.26 Er Erbium	69 168.93 Tm Thulium	70 173.04 Yb Ytterbium	71 174.97 Lu Lutetium
98 (261) Cf Californium	99 (262) Es Einsteinium	100 (267) Fm Fermium	101 (268) Md Mendelevium	102 (269) No Nobelium	103 (262) Lr Lawrencium

Elemental information

Each element is given a tile with important information to identify it. These include the element's name, symbol, atomic number, and atomic mass number.

Atomic number
This is the number of protons in the nucleus of an atom of an element. A beryllium atom has four protons in its nucleus, so its atomic number is four.

Name
The elements are named after people, places, their sources, and many other things.



Atomic mass number
This number shows the average mass of the atoms of all the naturally occurring forms (isotopes) of a given element. Beryllium has 12 known isotopes, and their average mass is 9.0122 AMU (atomic mass units).

Chemical symbol
Each element is given either a one- or two-letter symbol. This is the shortened version of the element's English or Latin name. The first letter is always uppercase, and when there are two letters, the second is always lowercase.



Dmitri Mendeleev

The Russian scientist Mendeleev is usually thought of as the pioneer of the modern periodic table. His original arrangement organized the elements by atomic mass and also left gaps that accurately predicted the existence of some unknown elements, which would be discovered later.